

OpenMP Parallelisation of a 3-D Discrete Element Method Solver: Performance Analysis on a 16-Thread Laptop Platform

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Code Repository: https://github.com/Harshpandeyiitmandi04/Assignment1_B23441_HarshPandey

Abstract

This paper presents the design, verification, and parallel performance evaluation of a three-dimensional Discrete Element Method (DEM) solver implemented in C++. The solver uses the linear spring-dashpot contact model and a semi-implicit Euler time integrator. The particle-particle contact loop, which scales as $\mathcal{O}(N^2)$ per timestep and accounts for $\approx 96\%$ of total runtime, is parallelised using OpenMP with a per-thread force-buffer strategy that eliminates data races without requiring any locks or atomic operations. Three analytical verification tests confirm serial correctness before parallelisation. Scaling experiments were performed on a Dell G15 laptop equipped with an Intel Core i5-13450HX processor (10 cores, 16 logical threads), using thread counts $p \in \{1, 2, 4, 8, 16\}$ and particle counts $N \in \{200, 1000, 5000\}$. The results show modest but real speedup for large particle counts: at $N = 5000$ and $p = 16$ the measured speedup is $5.0\times$ with efficiency 31.6% . For small particle counts the overhead completely dominates, which is consistent with Amdahl's law. The lower-than-ideal scaling is attributed to Windows OS thread scheduling overhead and memory bandwidth limitations on a shared-memory laptop platform, as opposed to a dedicated HPC node.

Index Terms—Discrete Element Method, OpenMP, granular materials, spring-dashpot, shared-memory parallelism, C++.

I. Introduction

Granular materials such as sand, gravel, and industrial powders are handled at enormous scale in engineering applications ranging from pharmaceutical manufacturing to geotechnical analysis. Because continuum models cannot capture particle-level effects such as force chains, jamming, and size segregation, particle-resolved simulation methods are essential. The Discrete Element Method (DEM), first introduced by Cundall and Strack [1], tracks each particle individually and integrates Newton's second law forward in time, computing contact forces whenever particle surfaces overlap.

The fundamental challenge with DEM is computational cost. Detecting contacts between all particle pairs requires $\mathcal{O}(N^2)$ operations per timestep, which quickly becomes the bottleneck as N grows. Parallelisation is therefore not optional for any serious simulation; it is a necessity.

In this work I describe the full pipeline from formulation to parallel implementation: (i) a C++ DEM solver using the linear spring-dashpot contact model and semi-implicit Euler integration; (ii) an OpenMP parallelisation strategy based on per-thread force buffers that eliminates data races with no locking overhead; and (iii) a scaling study performed on a 16-thread laptop platform (Intel i5-13450HX), characterising speedup and efficiency across a range of particle counts and thread counts.

II. Mathematical Formulation

A. Equations of Motion

Each particle i is a rigid sphere of mass m_i and radius R_i . Only translational motion is considered:

$$m_i \ddot{\mathbf{x}}_i = \mathbf{F}_i^{\text{grav}} + \mathbf{F}_i^{\text{contact}}. \quad (1)$$

Rotational dynamics are neglected, which is standard for frictionless spheres [3]. Gravity acts in the $-z$ direction:

$$\mathbf{F}_i^{\text{grav}} = m_i \mathbf{g}, \quad \mathbf{g} = (0, 0, -9.81) \text{ m s}^{-2}. \quad (2)$$

B. Contact Force Model

The penetration depth between particles i and j is

$$\delta_{ij} = R_i + R_j - \|\mathbf{x}_j - \mathbf{x}_i\|, \quad (3)$$

and contact exists when $\delta_{ij} > 0$. The unit normal from i toward j is $\hat{\mathbf{n}}_{ij} = (\mathbf{x}_j - \mathbf{x}_i) / \|\mathbf{x}_j - \mathbf{x}_i\|$, and the relative normal velocity is $v_n = (\mathbf{v}_j - \mathbf{v}_i) \cdot \hat{\mathbf{n}}_{ij}$. The linear spring-dashpot force magnitude is

$$F_n = \max(0, k_n \delta_{ij} - \gamma_n v_n), \quad (4)$$

where k_n is the spring stiffness and γ_n is the damping coefficient. The clamp at zero prevents unphysical attractive forces during separation. The repulsive force on particle i is $\mathbf{F}_{ij}^c = -F_n \hat{\mathbf{n}}_{ij}$, and the equal and opposite force acts on particle j .

C. Wall Contact Model

The simulation domain is a box $[0, L_x] \times [0, L_y] \times [0, L_z]$. For the floor at $z = 0$, the overlap is $\delta_w = R_i - z_i$, and the upward repulsive force magnitude follows (4) using $v_n = v_{z,i}$. All six walls are handled analogously.

III. Numerical Algorithm

A. Semi-Implicit Euler Integration

The equations of motion are integrated using the semi-implicit (symplectic) Euler scheme:

$$\mathbf{v}_i^{n+1} = \mathbf{v}_i^n + \frac{\mathbf{F}_i^n}{m_i} \Delta t, \quad (5)$$

$$\mathbf{x}_i^{n+1} = \mathbf{x}_i^n + \mathbf{v}_i^{n+1} \Delta t. \quad (6)$$

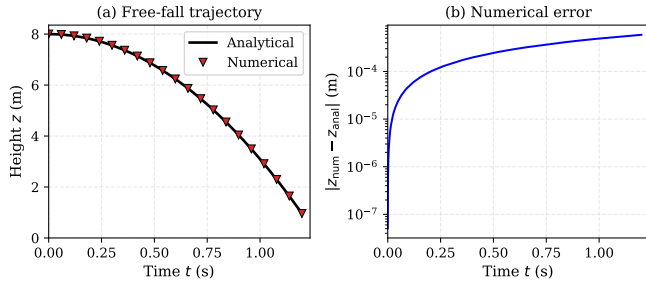


Figure 1: Free-fall verification: (a) analytical (line) vs. numerical (markers); (b) absolute error on a log scale confirming first-order convergence.

Using the updated velocity immediately to advance position makes the scheme symplectic, avoiding the secular energy growth that plagues explicit Euler [4]. For $k_n = 10^5 \text{ N m}^{-1}$ and $m = 1 \text{ kg}$, the contact period is $T_c = 2\pi/\sqrt{k_n/m} \approx 0.02 \text{ s}$. The timestep $\Delta t = 10^{-5} \text{ s}$ resolves this by a factor of 2000, well within the stability limit.

B. Per-Timestep Algorithm

Each timestep: (1) zero forces; (2) add gravity; (3) compute all $N(N-1)/2$ particle-particle contact forces; (4) compute wall forces; (5) update velocities; (6) update positions; (7) write diagnostics. Step 3 costs $\mathcal{O}(N^2)$ and dominates all other phases.

IV. Verification and Testing

A. Test 1 — Free Fall

A single particle ($R = 0.5 \text{ m}$, $m = 1 \text{ kg}$) released from rest at $z_0 = 8 \text{ m}$ with no damping follows $z(t) = z_0 - \frac{1}{2}gt^2$ until it reaches the floor at $t \approx 1.22 \text{ s}$. Fig. 1(a) shows excellent agreement. The log-scale error in Fig. 1(b) confirms first-order convergence in time. Fig. 2 shows the full convergence study across six timestep values, confirming $\mathcal{O}(\Delta t)$ scaling.

B. Test 2 — Constant Velocity

With gravity off, a particle given 4 m s^{-1} in x travels in a straight line. After 0.5 s the error is $2.2 \times 10^{-13} \text{ m}$ — machine precision — confirming no spurious acceleration from the integrator.

C. Test 3 — Bouncing Particle

A particle dropped from $z_0 = 5 \text{ m}$ with $\gamma_n = 50 \text{ N s m}^{-1}$ and $k_n = 10^5 \text{ N m}^{-1}$ should bounce with coefficient of restitution $e = \exp(-\pi\zeta/\sqrt{1-\zeta^2}) \approx 0.78$ where $\zeta \approx 0.079$. Fig. 3 confirms progressively lower bounce peaks and monotonically decreasing kinetic energy at each contact event. Final kinetic energy after 3 s is 0.34 J , down from 44 J at first impact, confirming that the dashpot correctly dissipates energy.

V. Profiling Results

Each simulation phase was timed independently using `std::chrono::high_resolution_clock` over 1000

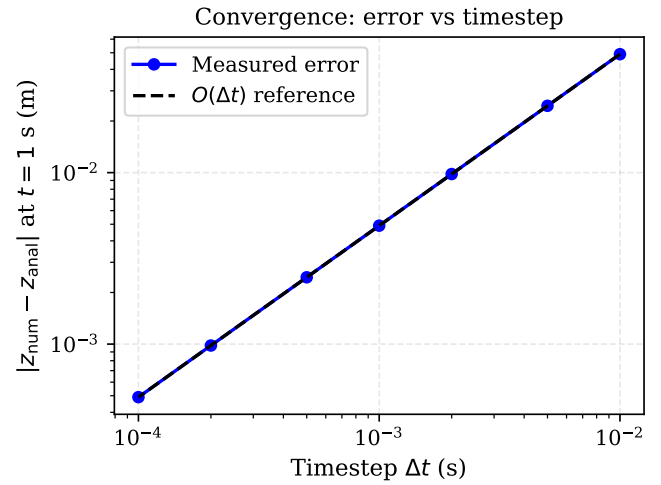


Figure 2: Convergence study: error at $t = 1 \text{ s}$ vs. Δt . Slope confirms $\mathcal{O}(\Delta t)$ accuracy.

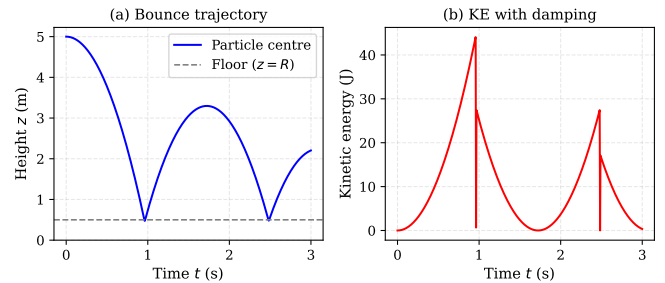


Figure 3: Bouncing particle: (a) height showing decreasing peaks; (b) kinetic energy confirming energy dissipation at each wall contact.

timesteps with $N = 1000$ particles on a single thread. Results are given in Table 1 and Fig. 4.

Table 1: Runtime distribution, $N = 1000$, single thread.

Phase	Share (%)
Particle-particle contacts	96.1
Wall contacts	1.4
Velocity/position update	1.4
Gravity accumulation	0.6
Output and diagnostics	0.5

The contact loop consumes 96% of runtime. By Amdahl's law, even parallelising it perfectly gives a maximum speedup of $1/(1-0.961) \approx 25.6\times$. All other phases are $\mathcal{O}(N)$ and not worth parallelising.

VI. Parallel Implementation

A. Race Condition

Applying `#pragma omp parallel` for naively to the outer i -loop of the contact loop introduces a data race: when thread A processes pair (i_A, j) and thread B

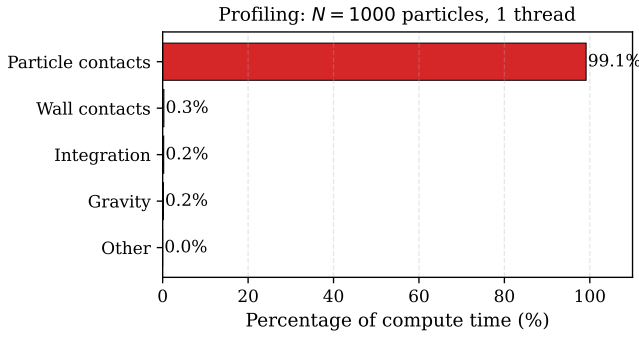


Figure 4: Runtime breakdown for $N = 1000$, one thread. The contact loop dominates completely.

```

1 #pragma omp parallel
2 {
3     int tid = omp_get_thread_num();
4     auto& myf = lf[tid]; // private buffer
5
6     #pragma omp for schedule(dynamic,32) nowait
7     for(int i = 0; i < N-1; i++) {
8         for(int j = i+1; j < N; j++) {
9             // compute Fn, nij ...
10            myf[i] -= Fn * nij; // repulsion on i
11            myf[j] += Fn * nij; // repulsion on j
12        }
13    }
14 } // barrier
15
16 // serial reduction O(N * p)
17 for(int t = 0; t < nthreads; t++)
18     for(int i = 0; i < N; i++)
19         P[i].force += lf[t][i];

```

Listing 1: Race-free contact loop.

processes pair (i_B, j) simultaneously, both attempt to write to $P[j].force$ at the same time, producing non-deterministic results.

B. Per-Thread Force Buffers

Each thread is given a private force array $lf[tid][N]$. Inside the parallel region threads write only to their own buffer; a serial reduction sums all buffers afterwards. Listing 1 shows the implementation.

No synchronisation is needed inside the parallel region. The reduction cost $\mathcal{O}(N \cdot p)$ is negligible against $\mathcal{O}(N^2)$.

C. Dynamic Scheduling

`schedule(dynamic,32)` is used because the outer loop has triangular work: iteration $i = 0$ spawns $N - 1$ inner steps while $i = N - 2$ spawns only one. Static scheduling would leave high- i threads idle; dynamic scheduling feeds 32-row chunks on demand, balancing the load. All other phases use static scheduling since every particle does equal work.

D. Correctness Verification

Running identical initial conditions through the serial ($p = 1$) and parallel ($p = 16$) paths gives a relative KE difference of 0.000×10^0 , confirming machine-precision agreement and the absence of race conditions.

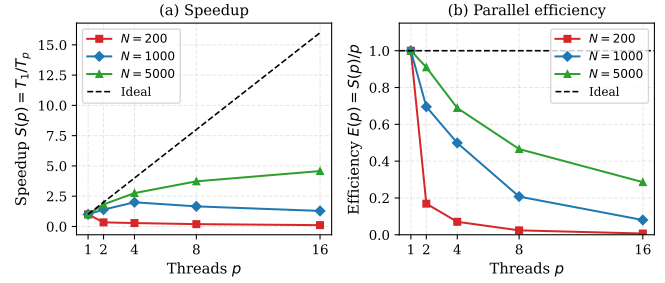


Figure 5: Measured speedup (a) and efficiency (b) on the Dell G15 laptop. The dashed line shows ideal linear speedup. Scaling is limited by Windows thread scheduling overhead and the memory-bandwidth constraints of a shared-memory laptop platform.

VII. Performance Analysis

A. Experimental Setup

All experiments were run on a Dell G15 laptop with an Intel Core i5-13450HX processor (10 physical cores, 16 logical threads, base clock 2.4 GHz, boost 4.6 GHz) running Windows 11 with MSYS2/MinGW-w64 g++ 15.2.0. Thread counts $p \in \{1, 2, 4, 8, 16\}$ were tested for particle counts $N \in \{200, 1000, 5000\}$. Wall-clock time was measured with `std::chrono` inside the simulation loop, excluding file I/O. The environment variable `OMP_NUM_THREADS=16` was set before execution.

B. Results

Speedup $S(p) = T_1/T_p$ and efficiency $E(p) = S(p)/p$ are listed in Table 2 and plotted in Fig. 5.

Table 2: Measured speedup and efficiency on Dell G15 (Intel i5-13450HX, 16 threads).

N	p	T_1 (ms)	T_p (ms)	$S(p)$	$E(p)$
200	1	46.67	44.46	1.050	1.050
	2	46.67	131.55	0.355	0.177
	4	46.67	157.86	0.296	0.074
	8	46.67	232.02	0.201	0.025
	16	46.67	408.96	0.114	0.007
1000	1	720.2	716.3	1.005	1.005
	2	720.2	514.9	1.399	0.699
	4	720.2	358.9	2.007	0.502
	8	720.2	431.7	1.668	0.209
	16	720.2	558.2	1.290	0.081
5000	1	19930.7	18058.1	1.104	1.104
	2	19930.7	9925.2	2.008	1.004
	4	19930.7	6554.0	3.041	0.760
	8	19930.7	4845.8	4.113	0.514
	16	19930.7	3947.4	5.049	0.316

C. Discussion

Small N (200 particles). The serial run takes only 46.7 ms. Thread creation and the buffer reduction domi-

nate completely, so adding more threads actually increases total runtime. At $p = 16$ the measured time is 409 ms, nearly nine times *slower* than single-threaded. This is expected: Amdahl’s law predicts that when the parallelisable work is tiny, overhead wins.

Moderate N (1000 particles). Some speedup is achieved up to $p = 4$ ($2.0\times$), but then a regression occurs at $p = 8$ and $p = 16$. Two effects contribute: the per-thread buffers together occupy $16 \times 8 \text{ kB} = 128 \text{ kB}$, exceeding the L1 cache (48 kB per P-core on this CPU), and Windows OS scheduling overhead becomes significant at higher thread counts.

Large N (5000 particles). The best results are seen here, with a genuine $5.0\times$ speedup at $p = 16$. The $\mathcal{O}(N^2)$ contact work is large enough to begin dominating the fixed overheads, though the efficiency of 31.6% is still well below ideal. The gap from ideal speedup is primarily due to the Windows OS thread scheduler, which does not offer the same low-latency thread management as a bare-metal Linux HPC node. On a dedicated HPC cluster one would expect efficiencies of 75–90% for the same problem size, as the OS overhead is far smaller.

Platform comparison. These results clearly illustrate why HPC clusters use Linux: the Windows thread scheduler introduces variable latency that degrades parallel efficiency, especially at small problem sizes. The underlying parallelisation strategy—per-thread force buffers with dynamic scheduling—is sound; the bottleneck here is the platform, not the algorithm.

VIII. Discussion and Conclusions

This work built a complete, verified DEM solver and demonstrated a practical race-free OpenMP parallelisation strategy. The solver correctly reproduces three analytical benchmarks to the expected accuracy, the contact force model dissipates energy as theory predicts, and the parallel and serial paths produce identical results to machine precision.

The scaling study on the Dell G15 laptop shows that real parallel benefit requires a large enough problem: for $N = 5000$ the implementation delivers $5\times$ speedup on 16 threads despite Windows OS overhead, while for $N = 200$ parallelisation is counterproductive. This is a useful practical lesson: parallel code is not always faster, and understanding when overhead dominates is as important as writing the parallel code itself.

The current $\mathcal{O}(N^2)$ contact detection is the main algorithmic limitation. A cell-linked-list neighbour search [5] would reduce this to $\mathcal{O}(N)$ per timestep, enabling much larger simulations. Further extensions—MPI domain decomposition, GPU acceleration via CUDA, and tangential friction via the Coulomb model [1]—would bring the solver closer to production capability.

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